



Indian Institute of Technology Guwahati
Department of Physics
PH102
Tutorial-4

Date: 30-01-2024

1. Suppose the electric field in some region is found to be $\vec{E} = kr^3\hat{r}$, in spherical coordinates (k is some constant).
 - (a) Find the charge density $\rho(\vec{r})$.
 - (b) Find the total charge contained in a sphere of radius R , centered at the origin. (Do it in two different ways.)
2. Find the electric field inside a sphere that carries a charge density proportional to the distance from the origin, $\rho = kr$, for some constant k . [Hint: This charge density is not uniform, and you must integrate to get the enclosed charge.]
3. A thick spherical shell carries charge density

$$\rho = \frac{k}{r^2}(a \leq r \leq b)$$

, (shown in fig 1). Find the electric field in the three regions: (i) $r < a$, (ii) $a < r < b$, (iii) $r > b$. Plot $|\vec{E}|$ as a function of r , for the case $b = 2a$

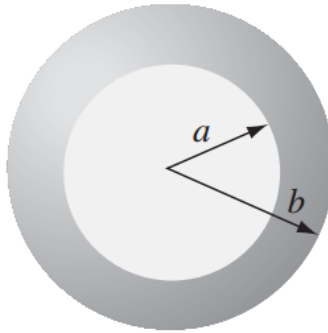


Figure 1

4. Find the potential inside and outside a uniformly charged solid sphere whose radius is R and whose total charge is q . Use infinity as your reference point. Compute the gradient of V in each region, and check that it yields the correct field. Sketch $V(r)$.
5. An infinite plane slab, of thickness $2d$, carries a uniform volume charge density ρ (shown in fig 2). Find the electric field, as a function of y , where $y = 0$ at the center. Plot E versus y , calling E positive when it points in the $+y$ direction and negative when it points in the $-y$ direction.

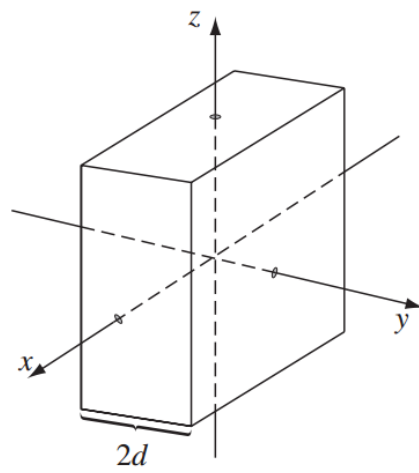


Figure 2

- 1) Suppose the electric field in some region is found to be $\vec{E} = kr^3 \hat{r}$, in spherical co-ordinates (k is some constant).
- (a) Find the charge density ρ .
- (b) Find the total charge contained in a sphere of radius R , centered at the origin. (Do it in 2 different ways)

Ans Gauss law: $\oint \vec{E} \cdot d\vec{s} = \frac{Q_{enc}}{\epsilon_0} \quad \text{--- (1)}$

Gauss divergence theorem, $\oint_S \vec{E} \cdot d\vec{s} = \int_V (\vec{\nabla} \cdot \vec{E}) d\tau \quad \text{--- (2)}$

but $Q_{enc} = \int_V \rho d\tau \quad \text{--- (3)}$

using (3) in (1), $\oint \vec{E} \cdot d\vec{s} = \frac{1}{\epsilon_0} \int \rho d\tau \quad \text{--- (4)}$

Comparing (2) & (4), $\int (\vec{\nabla} \cdot \vec{E}) d\tau = \int \frac{\rho}{\epsilon_0} d\tau$

$\Rightarrow \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

$\Rightarrow \rho = \epsilon_0 (\vec{\nabla} \cdot \vec{E})$

Given $\vec{E} = kr^3 \hat{r}$

$\rho = \epsilon_0 \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \cdot kr^3) = \epsilon_0 \frac{1}{r^2} k (5r^4) = 5\epsilon_0 kr^2$

$\boxed{\rho = 5\epsilon_0 kr^2}$

(b) By Gauss Law, $Q_{enc} = \epsilon_0 \oint \vec{E} \cdot d\vec{s} = \epsilon_0 (\vec{E} \cdot \vec{A})$

$Q_{enc} = \epsilon_0 (kR^3) (4\pi R^2)$

$\boxed{Q_{enc} = 4\pi \epsilon_0 kR^5}$

Also, $Q_{enc} = \int \rho d\tau = \int_0^R (5\epsilon_0 kr^2) (4\pi r^2 dr)$

$= 4\pi \epsilon_0 k \frac{R^5}{5} = 4\pi \epsilon_0 kR^5$

$\boxed{Q_{enc} = 4\pi \epsilon_0 kR^5}$

[Concept of flux
 $\phi = \frac{q}{\epsilon_0}$

$\phi = \vec{E} \cdot \vec{A}$

$\Rightarrow \frac{q}{\epsilon_0} = \vec{E} \cdot \vec{A}$

$q = \epsilon_0 (\vec{E} \cdot \vec{A})$

- 1) Find the electric field inside a sphere that carries a charge density proportional to the distance from the origin, $\rho = kr$, for some constant k .

[Hint: This charge density is not uniform & you must integrate to get the enclosed charge.]

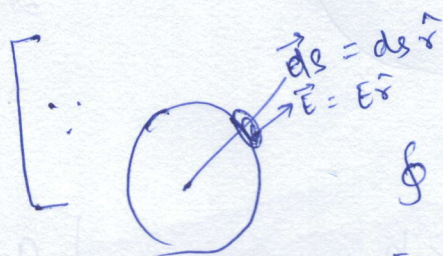
Ans

A. T Gauss Law,



$$\oint \vec{E} \cdot d\vec{s} = E \cdot 4\pi r^2$$

for spherical surface



$$\begin{aligned} \oint \vec{E} \cdot d\vec{s} &= \oint E \hat{r} \cdot ds \hat{r} \\ &= \oint E ds \\ &= E \oint ds \end{aligned}$$

$$\oint \vec{E} \cdot d\vec{s} = E \cdot 4\pi r^2$$

$$\Rightarrow E \cdot 4\pi r^2 = \frac{1}{\epsilon_0} Q_{enc}$$

$$= \frac{1}{\epsilon_0} \int \rho dz$$

$$= \frac{1}{\epsilon_0} \int_0^r (kr) r^2 \sin\theta d\theta d\phi dr$$

$$= \frac{1}{\epsilon_0} k \cdot 4\pi \int_0^r r^3 dr$$

$$= \frac{4\pi k}{\epsilon_0} \left[\frac{r^4}{4} \right]_0^r = \frac{4\pi k r^4}{4\epsilon_0}$$

$$\Rightarrow E \cdot 4\pi r^2 = \frac{4\pi k r^4}{4\epsilon_0}$$

$$\Rightarrow \boxed{\vec{E} = \frac{kr^2}{\epsilon_0} \hat{r}}$$

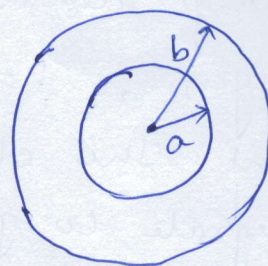
$$\boxed{\oint \vec{E} \cdot d\vec{s} = E(4\pi r^2)}$$

3) A thick spherical shell carries charge density $\rho = \frac{k}{r^2}$ ($a \leq r \leq b$)

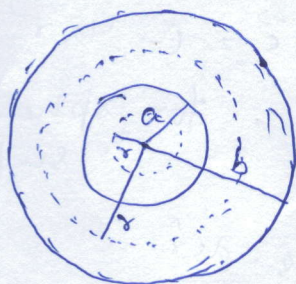
Find the electric field in 3 regions:

(i) $r < a$ (ii) $a < r < b$ (iii) $r > b$

Plot $|E|$ as a function of r , for the case $b = 2a$.



Ans



(a) Inside ($r < a$)

$$\oint \vec{E} \cdot d\vec{s} = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow q_{enc} = 0$$

$$\Rightarrow \boxed{\vec{E} = 0}$$

$$\left[\begin{aligned} & \int \frac{k}{r^2} \sin \theta dr d\theta d\phi \\ & = k \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \int_a^r dr \\ & = k(2)(2\pi) \int_a^r dr \end{aligned} \right]$$

(b) $a < r < b$

$$\oint \vec{E} \cdot d\vec{s} = \frac{q_{enc}}{\epsilon_0}$$

$$q_{enc} = \int \rho d\tau$$

$$= 4\pi k \int_a^r \frac{1}{r^2} r^2 dr$$

$$= 4\pi k \int_a^r dr = 4\pi k(r-a)$$

$$\Rightarrow E(4\pi r^2) = \frac{4\pi k(r-a)}{\epsilon_0}$$

$$\Rightarrow \boxed{\vec{E} = \frac{k(r-a)}{\epsilon_0 r^2} \hat{r}}$$

(c) $r > b$

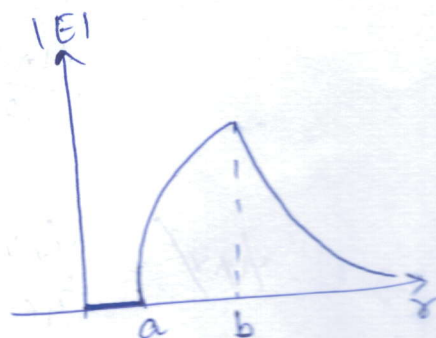
$$q_{enc} = 4\pi k \int_a^b dr = 4\pi k(b-a)$$

$$\Rightarrow \oint \vec{E} \cdot d\vec{s} = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow E \cdot 4\pi r^2 = \frac{4\pi k(b-a)}{\epsilon_0}$$

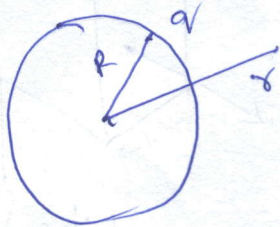
$$\Rightarrow \boxed{\vec{E} = \frac{k(b-a)}{\epsilon_0 r^2} \hat{r}}$$

Plot for $b = 2a$



Q Find the potential inside and outside a uniformly charged solid sphere whose radius is R & whose total charge is q . Use infinity as your reference point. Compute the gradient of V in each region and check that it yields the correct field. Sketch $V(r)$

Ans (a) V_{outside}



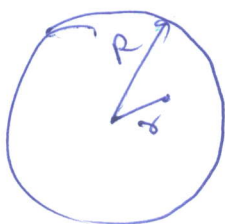
$$V = - \int_{\infty}^r E \cdot dr$$

$$\Rightarrow V_{\text{out}} = - \int_{\infty}^r \frac{kq}{r^2} dr$$

$$\Rightarrow V_{\text{out}} = \left[\frac{kq}{r} \right]_{\infty}^r = \frac{kq}{r}$$

$$\boxed{V_{\text{out}} = \frac{kq}{r}}$$

(b) V_{inside}



$$V = - \int_{\infty}^r E dr = - \int_{\infty}^R E_{\text{out}} dr - \int_R^r E_{\text{in}} dr$$

$$\Rightarrow V_{\text{in}} = - \int_{\infty}^R \frac{kq}{r^2} dr - \int_R^r \frac{kq r}{R^3} dr$$

$$= \left[\frac{kq}{r} \right]_{\infty}^R - \left[\frac{kq r^2}{2R^3} \right]_R^r$$

$$= \frac{kq}{R} - \left[\frac{kq r^2}{2R^3} - \frac{kq R^2}{2R^3} \right]$$

$$\Rightarrow V_{in} = \frac{kq}{R} - \frac{kq}{R} \left[\frac{r^2}{2R^2} - \frac{1}{2} \right]$$

$$= \frac{kq}{R} \left[1 - \frac{r^2}{2R^2} + \frac{1}{2} \right]$$

$$\Rightarrow V_{in} = \frac{kq}{R} \left[\frac{3}{2} - \frac{r^2}{2R^2} \right] \Rightarrow \boxed{V_{in} = \frac{3}{2} \frac{kq}{R} \left[1 - \frac{r^2}{3R^2} \right]}$$

When $r > R$, $\vec{\nabla} V = \frac{q}{4\pi\epsilon_0} \frac{\partial}{\partial r} \left(\frac{1}{r} \right) \hat{r} = -\frac{q}{4\pi\epsilon_0} \frac{1}{r^2} \hat{r}$

$$\Rightarrow \vec{E} = -\vec{\nabla} V = \frac{q}{4\pi\epsilon_0} \frac{1}{r^2} \hat{r}$$

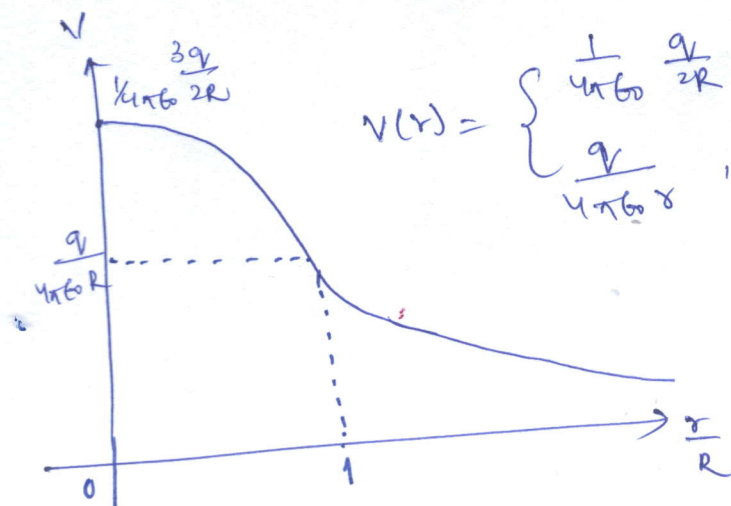
When $r < R$, $\vec{\nabla} V = \frac{q}{4\pi\epsilon_0} \frac{1}{2R} \frac{\partial}{\partial r} \left(3 - \frac{r^2}{R^2} \right) \hat{r}$

$$= \frac{q}{4\pi\epsilon_0} \frac{1}{2R} \left(-\frac{2r}{R^2} \right) \hat{r} = -\frac{q}{4\pi\epsilon_0} \frac{r}{R^3} \hat{r}$$

$$\text{So, } \vec{E} = -\vec{\nabla} V = \frac{1}{4\pi\epsilon_0} \frac{q}{R^3} r \hat{r}$$

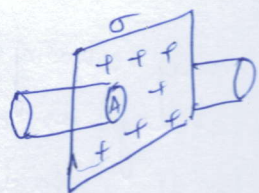
$$V(r) = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{q}{2R} \left[3 - \left(\frac{r}{R} \right)^2 \right] & \text{if } r < R \\ \frac{q}{4\pi\epsilon_0 r} & \text{if } r > R \end{cases}$$

Plot
 $V(r) \propto \frac{r}{R}$



5) An infinite plane slab of thickness $2d$, carries a uniform volume charge density ρ (Fig.). Find the electric field, as a function of y , where, $y=0$ at the center. Plot E vs y , calling E positive when it points in the $+y$ dirⁿ & negative when it points in the $-y$ direction.

Solⁿ. The Electric field due to a infinite sheet of charge we can get by using a pill box: Gaussian surface



$$\oint \vec{E} \cdot d\vec{s} = \frac{q_{enc}}{\epsilon_0}$$

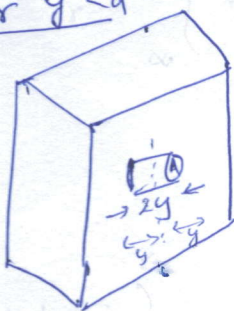
$$E(2A) = \frac{\sigma A_{enc}}{\epsilon_0}$$

$$\Rightarrow E(2A) = \frac{\sigma A}{\epsilon_0} \Rightarrow E = \frac{\sigma}{2\epsilon_0}$$

Charge is distributed infinitely along x - z plane so $\vec{E}$ will be a funcⁿ of y .

Let's construct a pill box whose boundaries are $\perp$ & $\parallel$ to the xz plane.

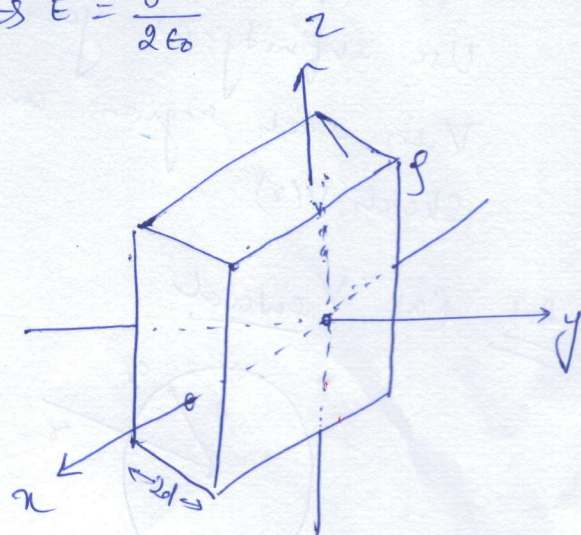
a) for $y < d$



$$\oint \vec{E} \cdot d\vec{s} = \frac{q_{enc}}{\epsilon_0}$$

$$\Rightarrow E(2A) = \frac{\rho A y}{\epsilon_0}$$

$$\Rightarrow E = \frac{\rho y}{\epsilon_0}$$



(Fig)

$$\vec{E} = \frac{\rho y}{\epsilon_0} \hat{y}$$

b) for $y > d$

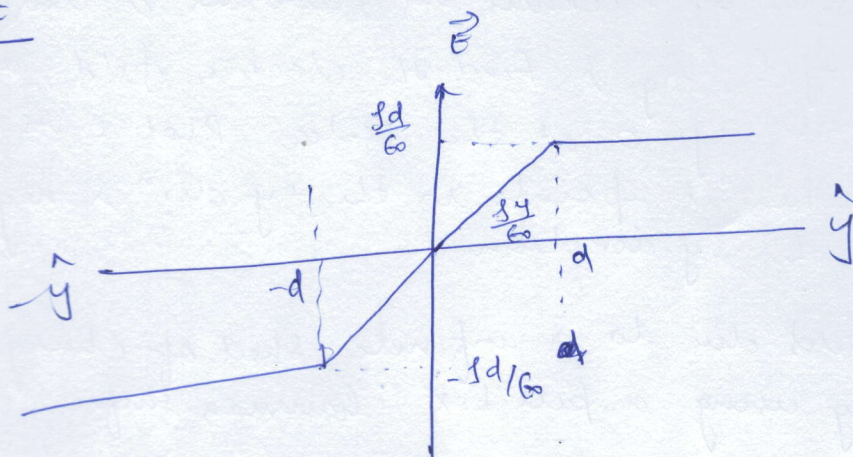
$$\oint \vec{E} \cdot d\vec{s} = \frac{q_{enc}}{\epsilon_0}$$

$$E(2A) = \frac{\rho A 2d}{\epsilon_0}$$

$$\Rightarrow E = \frac{\rho d}{\epsilon_0}$$

$$\Rightarrow \vec{E} = \frac{\rho d}{\epsilon_0} \hat{y}$$

Plot



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